CM - formula Sheet :-

UNIT-01

Taylor's Series

$$y' = \frac{dy}{dx} = f(x,y)$$

$$y = y_0 + xy_0' + \frac{x^2 y_0''}{2!} + \frac{x^3 y_0'''}{3!} + \cdots$$

Rolle's Theorem

(i) $f(x)$ is Continous in $[a,b]$
(ii) $f(x)$ is differentiable in (a,b)
(iii) $f(a) = f(b)$

then $\exists$ atleast one point $c \in (a,b)$ so that $f'(c) = 0$

Mean Value theorem

(i)
(ii) same Rolle's theorem
(iii) $f(a) \neq f(b)$

then $\exists c \in (a,b)$ so that

$$f'(c) = \frac{f(b) - f(a)}{b-a}$$

Bisection Method

Let $f(x) = 0$

Suppose, $f(0) = -ve$
$f(1) = -ve$ } root
$f(2) = +ve$ } lie b/w them

1st Approx.

$x_1 = \frac{1+2}{2} = 1.5$; $f(1.5) = -ve$

2nd Approx.

$x_2 = \frac{1.5+2}{2} = 1.75$; $f(1.75) = +ve$

3rd Approx.

$x_3 =$ - - - - - - - -

So on till required accuracy.

Secant Method

Let $f(x) = 0$

Suppose, $f(0) = -ve$
$f(1) = -ve \rightarrow x_0$
$f(2) = +ve \rightarrow x_1$

$$x_{n+1} = \frac{x_{n-1} f(x_n) - x_n f(x_{n-1})}{f(x_n) - f(x_{n-1})}$$

$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)} \quad ; \quad f(x_2) =$$

$x_3 =$ similarly -

Newton Raphson Method

Let $f(x) = 0$

Suppose, $f(0) = -ve$
$f(1) = -ve$ } root lies
$f(2) = +ve$ } b/w them

Let $x_0 = 1$ or 2 or $\frac{1+2}{2} = 1.5$

also find $f'(x)$

$$x_n = x_{n-1} - \frac{f(x_{n-1})}{f'(x_{n-1})}$$

$n=1$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Absolute Error :

If x is true value of x' its Approximate value.

$E_a = |x - x'|$

Relative Error :

$$E_R = \left| \frac{x - x'}{x} \right|$$

% Error :

$$E_p = \left| \frac{x - x'}{x} \right| \times 100$$

Inherent errors :

Errors which are already present in the statement of a problem before its solution.

Rounding errors :

Errors which arise from the process of rounding off the numbers.

Truncation errors :

Errors caused by using Approximate results.

! Do it Yourself :)

- fibonacci
- Golden Search
- Newton's Method
- Steepest
- Nelder-Mead Algo.

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UNIT-02

Interpolation

→ Newton forward :

$$f(x+hu) = f(x) + \frac{u}{1!}\Delta f(x) + \frac{u(u-1)}{2!}\Delta^2 f(x) + \frac{u(u-1)(u-2)}{3!}\Delta^3 f(x) + \text{----}$$

(x: initial/Base, h: Interval)

→ Newton Backward :

$$f(x+hu) = f(x) + u\nabla f(x) + \frac{u(u+1)}{2!}\nabla^2 f(x) + \frac{u(u+1)(u+2)}{3!}\nabla^3 f(x) + \text{---}$$

$u = \frac{x - x_0}{h}$ (Base: initial)

Lagrange's interpolation for unequal interval :

Ex:-

x	5	6	9	11
y	12	13	14	16

$$f(x) = \frac{(x-6)(x-9)(x-11)}{(5-6)(5-9)(5-11)} \times 12 + \frac{(x-5)(x-9)(x-11)}{(6-5)(6-9)(6-11)} \times 13 + \frac{(x-5)(x-6)(x-11)}{(9-5)(9-6)(9-11)} \times 14 + \frac{(x-5)(x-6)(x-9)}{(11-5)(11-6)(11-9)} \times 16$$

Newton Divided Difference :

x	f(x)	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
x_0 5	12	$\frac{13-12}{6-5} = 1$	$\frac{\frac{1}{3}-1}{9-5} = -\frac{1}{6}$	$\frac{\frac{2}{15}-(-\frac{1}{6})}{11-5} = \frac{1}{20}$
x_1 6	13	$\frac{14-13}{9-6} = \frac{1}{3}$	$\frac{1-\frac{1}{3}}{11-6} = \frac{2}{15}$	
x_2 9	14	$\frac{16-14}{11-9} = 1$		
x_3 11	16			

given : x = 10

$$f(x) = f(x_0) + (x-x_0)\Delta f(x_0) + (x-x_0)(x-x_1)\Delta^2 f(x_0) + (x-x_0)(x-x_1)(x-x_2)\Delta^3 f(x_0) + \text{---}$$

Numerical integration :

$$I = \int_a^b f(x)\,dx \quad ; \quad h = \frac{b-a}{n}$$

① Trapezoidal Rule :-

$$\int_a^b f(x)\,dx = \frac{h}{2}\left[(y_0+y_n) + 2(y_1+y_2+\text{---}y_{n-1})\right]$$

∵ for any no. of interval.

② Simpson $\frac{1}{3}$ Rule : (Even interval)

$$\int_a^b f(x)\,dx = \frac{h}{3}\left[(y_0+y_n) + 4(y_1+y_3+\text{--}) + 2(y_2+y_4+y_6+\text{--})\right]$$

③ Simpson $\frac{3}{8}$ Rule : (for multiple of 3)

$$\int_a^b f(x)\,dx = \frac{3h}{8}\left[(y_0+y_n) + 3(y_1+y_2+y_4+\text{--}) + 2(y_3+y_6+y_9+\text{---})\right]$$

Gauss quadature formula

Qus → $I = \int_a^b f(x)\,dx$

Step ① : change interval [a,b] into [-1, 1] using $x = \frac{b-a}{2}t + \frac{b+a}{2}$

Step ② : Substitute x in f(x) integrand.

Step ③ : formula -

one-point (n=0) : $\int_{-1}^{1} f(x)\,dx = 2f(0)$

two-point : $\int_{-1}^{1} f(x)\,dx = f\left(\frac{-1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right)$

three-point : $\int_{-1}^{1} f(x)\,dx = \frac{1}{9}\left[5f\left(-\sqrt{\frac{3}{5}}\right) + 8f(0) + 5f\left(\sqrt{\frac{3}{5}}\right)\right]$

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UNIT - 03

Gauss Elimination Method.

Consider the system of eq's.

$$a_{11}x + a_{12}y + a_{13}z = b_1$$
$$a_{21}x + a_{22}y + a_{23}z = b_2$$
$$a_{31}x + a_{32}y + a_{33}z = b_3$$

Convert in Matrix form $AX = B$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

Make Augmented matrix, $C = [A : B]$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} : b_1 \\ a_{21} & a_{22} & a_{23} : b_2 \\ a_{31} & a_{32} & a_{33} : b_3 \end{bmatrix}$$

∵ use Row transformations,

$$C = \begin{bmatrix} - & - & - : - \\ 0 & - & - : - \\ 0 & 0 & - : - \end{bmatrix}$$

Convert back to system of eqs.

$$c_{11}x + c_{12}y + c_{13}z = d_1$$
$$c_{22}y + c_{23}z = d_2$$
$$c_{33}z = d_3$$

Now, solve these to get the Answers ☺.

Gauss Jordan

Similar to Gauss Elimination

$$\begin{bmatrix} 1 & 0 & 0 : - \\ 0 & 1 & 0 : - \\ 0 & 0 & 1 : - \end{bmatrix} \quad x = d_1, \; y = d_2, \; z = d_3$$

Eigen Value Problem : Power method.

Ex.

$$AX_0 = \begin{bmatrix} 3 & -5 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} -1 \\ +1 \end{bmatrix} \Rightarrow 2X_1$$

(Eigen value = 2, Eigen vector X_1)

$$AX_1 = \begin{bmatrix} 3 & -5 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} +8 \\ -6 \end{bmatrix} = 8 \begin{bmatrix} 1 \\ -0.75 \end{bmatrix} = 8X_2$$

(8 = bigger one)

$$AX_2 = \begin{bmatrix} 3 & -5 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ -0.75 \end{bmatrix} = \ldots X_3$$

So on till we get same value of successive iteration

LU Decomposition Methods.

Doolittle :

① $AX = B$

② $A = LU$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \quad U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

③ $LY = B$ ∵ $L\underbrace{UX}_{Y} = B$

where, $Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$ Solve to find y_1, y_2, y_3

④ $UX = Y$

from this we can find X (x, y, z) req. Sol. ☺

Crout's :

Same Procedure as Doolittle

$$L = \begin{bmatrix} a & 0 & 0 \\ b & c & 0 \\ d & e & f \end{bmatrix} \quad U = \begin{bmatrix} 1 & g & h \\ 0 & 1 & i \\ 0 & 0 & 1 \end{bmatrix}$$

Cholesky's : A → Matrix.

Conditions { → A is Symmetric i.e $A^t = A$
→ A is positive definite.

① $AX = B$

② $A = LL^T$

$$A = \begin{bmatrix} a & 0 & 0 \\ b & c & 0 \\ d & e & f \end{bmatrix} \begin{bmatrix} a & b & d \\ 0 & c & e \\ 0 & 0 & f \end{bmatrix} = \begin{bmatrix} a^2 & ab & ad \\ ab & b^2+c^2 & bd+ce \\ ad & bd+ec & d^2+e^2+f^2 \end{bmatrix}$$

(L, L^T)

③ $LY = B$, where $Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$

④ $L^T X = Y$

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Unit-04

Picard's Method.

$$y = y_0 + \int_{x_0}^{x} f(x,y)\,dx$$

I^st Approx. —

$$y_1 = y_0 + \int_{x_0}^{x} f(x, y_0)\,dx$$

2nd Approx. —

$$y_2 = y_0 + \int_{x_0}^{x} f(x, y_1)\,dx$$

Doing this till the desired degree of accuracy obtained.

Taylor's Series ($f(x,y) = y = \frac{dy}{dx}$)

$$y_1 = y_0 + hy_0' + \frac{h^2}{2!}y_0'' + \frac{h^3}{3!}y_0''' + \cdots$$

$h = (x - x_0)$

$x = x_0 + h$

Similarly

$$y_2 = y_1 + hy_1' + \frac{h^2}{2!}y_1'' + \frac{h^3}{3!}y_1''' + \cdots$$

and so on---

Euler's Method

Consider, $\frac{dy}{dx} = f(x,y)$; $y(x_0) = y_0$

To find : $y(x_n) = y_n$ (Initial)

Acc. to Euler's Method,

$$y_n = y_{n-1} + h \cdot f(x_{n-1}, y_{n-1}), \quad n = 1,2,3\cdots \quad \leftarrow \text{eqn. (1)}$$

h ← width of differencing

$$h = \frac{x_n - x_0}{n} = x_n - x_{n-1} = x_1 - x_0$$

n ← no. of interval. (Let any $n = 5, 10$, etc)

Put $n=1$, $y_1 = y_0 + hf(x_0, y_0)$

Put $n=2$, $y_2 = y_1 + hf(x_1, y_1)$

and so on.

from Euler's Method, Starting from y_0 and using eqn (1), we can get $y_1, y_2, y_3, \cdots y_n$.

Modified Euler's Method. ($x = x_0 + h$)

Step ①:- $y_1^{(0)} = y_0 + hf(x_0, y_0)$ (Euler Method)

$$y_1^{(n+1)} = y_0 + \frac{h}{2}\left[f(x_0,y_0) + f(x_1, y_1^{(n)})\right] \quad \text{(Modified)}$$

Put $n = 0, 1, 2, 3, \cdots$

Step ②: 1st Approx. $y_1^{(1)} = y_0 + \frac{h}{2}\left[f(x_0,y_0) + f(x_1, y_1^{(0)})\right]$

Similarly.

Step ③: 2nd Approx. $y_1^{(2)} = y_0 + \frac{h}{2}\left[f(x_0,y_0) + f(x_1, y_1^{(1)})\right]$

and so on. till required accuracy.

• if two consecutive values of $y_1^{(k)}$ & $y_1^{(k+1)}$ are same then $y_1 = y_1^{(k)}$

Preceding in the same manner, $y_2, y_3, \cdots$ are calculated.

Runge-Kutta (Second Order)

($x = x_0 + h$)

$\frac{dy}{dx} = f(x,y)$; $y(x_0) = y_0$

To find $y(x_1)$

$$k_1 = hf(x_n, y_n)$$

$$k_2 = hf(x_n + h, y_n + k_1)$$

$$y_{n+1} = y_n + \frac{1}{2}(k_1 + k_2)$$

Runge-Kutta (4th Order)

$$k_1 = hf(x_n, y_n)$$

$$k_2 = hf\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$$

$$k_3 = hf\left(x_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right)$$

$$k_4 = hf(x_n + h, y_n + k_3)$$

then Compute ;

Weighted mean of k_1, k_2, k_3 & k_4 → $k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$

$$y_{n+1} = y_n + k$$

Adams-Bashforth Predictor Corrector Method :

Consider, $\frac{dy}{dx} = f(x,y)$ with initial $y(x_0) = y_0$

To find : $y(x_n)$ → must be 4 starting Values of y.

find $y(x_1), y(x_2), y(x_3)$ using any Methods.

Then Calculate — $f_0 = f(x_0, y_0)$, f_1, f_2, f_3

Predictor → $y_4 = y_3 + \frac{h}{24}(55f_3 - 59f_2 + 37f_1 - 9f_0)$

find $f_4 = f(x_4, y_4)$

Corrector → $y_4 = y_3 + \frac{h}{24}(9f_4 + 19f_3 - 5f_2 + f_1)$

again find f_4 & again corrector.

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Milne Predictor-Corrector Method:

$\frac{dy}{dx} = f(x,y)$ with initial Cond^n $y(x_0) = y_0$

To find $y(x_0)_{+4}$

Find $y(x_1), y(x_2), y(x_3)$ using any Method!

Then Calculate:

$f_0 = f(x_0, y_0)$ $\quad f_1 = f(x_1, y_1)$

$f_2 = f(x_2, y_2)$ $\quad f_3 = f(x_3, y_3)$

Milne's Corrector formula:

$$y_4 = y_2 + \frac{h}{3}(f_2 + 4f_3 + f_4)$$

Find f_4 and again Apply Milne Method to find better value of y_4, we repeat this step until y_4 remains unchanged.

Numerical Solution of Partial Diff. eq^n.
(Parabolic, Hyperbolic, elliptical)

$$a\frac{\partial^2\phi}{\partial x} + b\frac{\partial^2\phi}{\partial x\,\partial y} + c\frac{\partial^2\phi}{\partial y^2}$$

$D = b^2 - 4ac$

$D > 0$ Hyperbola, 2 charac^n Sol^n

$D = 0$ Parabola, "

$D < 0$ Ellipse. No Sol^n.

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